

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - III

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Class :SY BSC IT	Batch :
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Stack Application:

Write a program to:

- a. Implement a stack using an array.

CODE:

```
prac 3 ds.py - C:/Users/Janhvi/AppData/Local/Programs/Python/Python314/python programs/prac 3 ds.py (3.14.5)
File Edit Format Run Options Window Help

import array as arr

stack1 = arr.array('i')

si = int(input("enter size of stack: "))
print(stack1)

def push():
    print("Push Operation:")
    if (len(stack1) == si):
        print("Stack overflow")
    else:
        ele = int(input("Enter element: "))
        stack1.append(ele)
        print(stack1)

def pop():
    print("Pop Operation")
    if (isEmpty() == 0):
        print("Stack Underflow")
    else:
        stack1.pop()
        print(stack1)

def peek():
    print("Peek Operation")
    if (isEmpty() == 0):
        print("Stack is empty")
    else:
        stack1.pop()
        print(stack1[-1])

def isEmpty():
    if len(stack1) == 0:
        return 0

num = 0
while (num < 5):
    print("Select any one program: \n1.Push \n2.Pop \n3.Peek \n4.isEmpty \n5.Exit")
    num = int(input("Select Choice: "))

    if (num == 1):
        push()
    elif (num == 2):
        pop()
    elif (num == 3):
        peek()
    elif (num == 4):
        isEmpty()
    else:
        print("Program terminated")
        print("Aditya Sopte S040")
```

OUTPUT:

PUSH OPERATION:

```
enter size of stack: 3
array('i')
Select any one program:
1.Push
2.Pop
3.Peek
4.isEmpty
5.Exit
Select Choice: 1
Push Operation:
Enter element: 23
array('i', [23])
Select any one program:
1.Push
2.Pop
3.Peek
4.isEmpty
5.Exit
Select Choice: 1
Push Operation:
Enter element: 24
array('i', [23, 24])
Select any one program:
1.Push
2.Pop
3.Peek
4.isEmpty
5.Exit
Select Choice: 1
Push Operation:
Enter element: 25
array('i', [23, 24, 25])
Select any one program:
1.Push
2.Pop
3.Peek
4.isEmpty
5.Exit
```

pop:

```
Select any one program:
1.Push
2.Pop
3.Peek
4.isEmpty
5.Exit
Select Choice: 2
Pop Operation
array('i', [23, 24])
```

peek:

```
Select any one program:  
1.Push  
2.Pop  
3.Peek  
4.isEmpty  
5.Exit  
Select Choice: 3  
Peek Operation  
23
```

isEmpty and isExit:

```
Select any one program:  
1.Push  
2.Pop  
3.Peek  
4.isEmpty  
5.Exit  
Select Choice: 4  
Select any one program:  
1.Push  
2.Pop  
3.Peek  
4.isEmpty  
5.Exit  
Select Choice: 5  
Program terminated  
Aditya Sopte S040  
>>>
```

- b. Convert an infix expression to postfix notation using a stack.(omit)

Curiosity Questions:

1. What's the difference between `peek()` and `pop()` in stack operations?
2. Why do we call a stack a "LIFO" structure? Can you think of a real-life example of this?
3. Can we store more than one data type (like `int` and `string`) in a stack using array module? Why or why not?
4. What happens if we try to `peek()` into an empty stack?
5. How can we check if the stack is empty before using `pop()`?